

Acknowledgement of Country



I would like to acknowledge the Bedegal people as the Traditional Custodians of this land. I would also like to pay my respects to the Elders, past and present, and extend that respect to other Aboriginal and Torres Strait Islanders who are present here today.

SCIF2001 – WORKSHOP 2

Literature Reviews

TODAY'S PLAN

Literature Reviews:

- 1) Why would you review literature?
- 2) How might you find literature to review?
- 3) What should you do with the literature you find?

LITERATURE REVIEW

What are you looking for, and how do you find it?

WHAT'S A LITERATURE REVIEW?

Literature refers to the existing body of scientific knowledge on a topic.

In a **literature review**, a Scholar (you) access the literature to achieve some goals.

Goals could be:

- 1) Learn more about a topic.
- 2) Find out if something's been done before.
- 3) Compare different approaches or techniques.
- 4) Identify sources of disagreement – opportunities for your own science!

WHAT'S A LITERATURE REVIEW?

What do all of these goals have in common?

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They're all about synthesising information!

A literature review is generally focused on building connections between papers and understanding your project's place in the world

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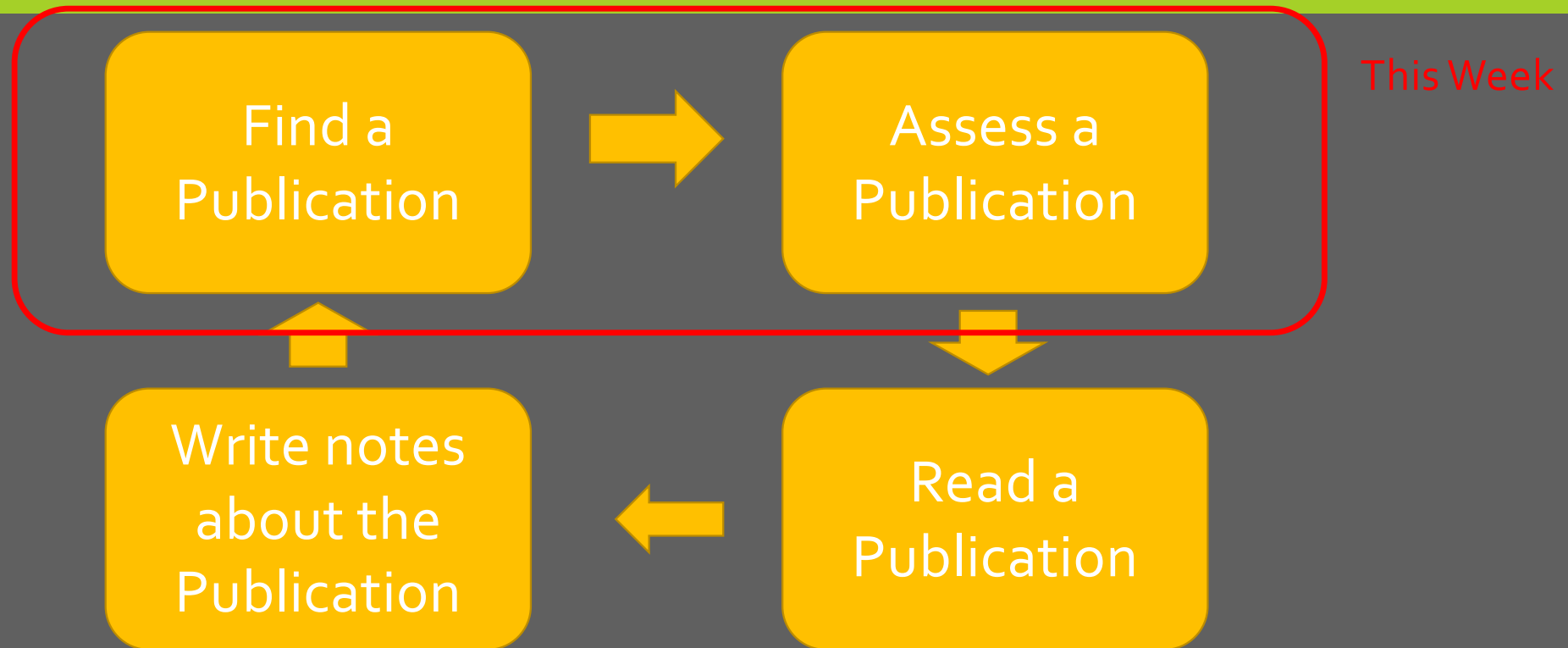
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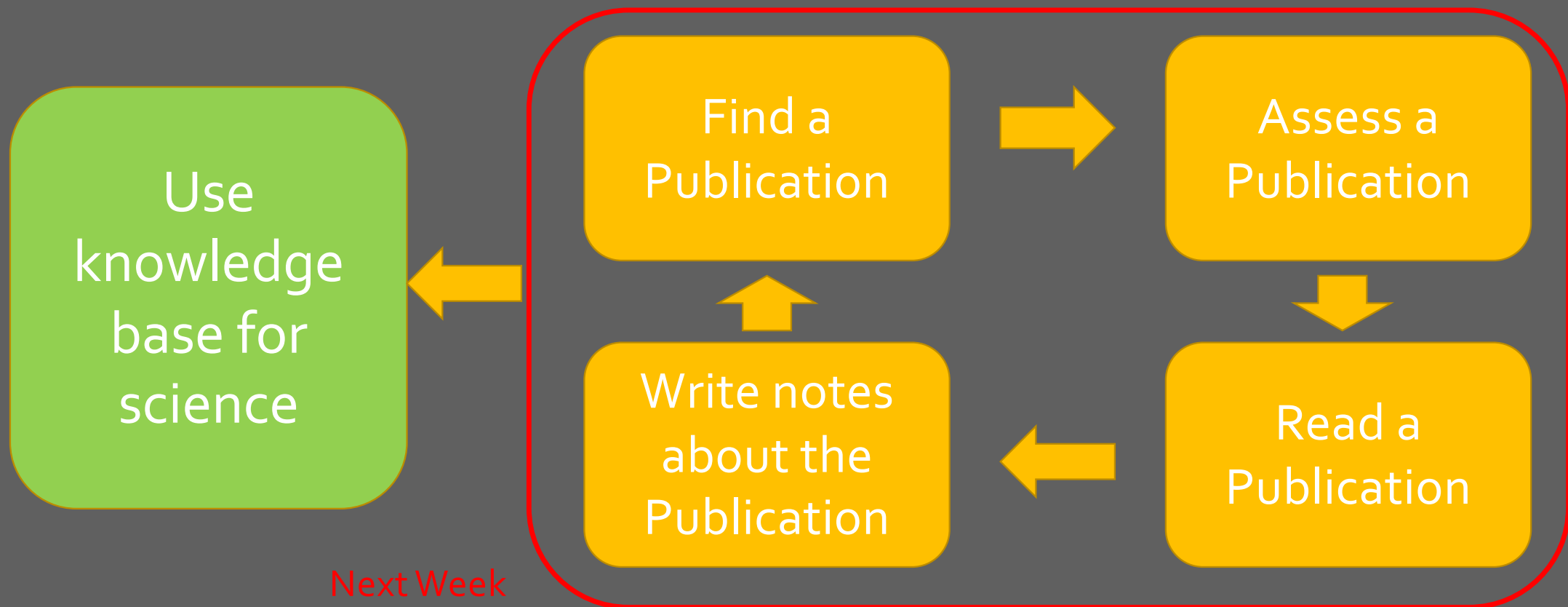
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ASSEMBLING YOUR KNOWLEDGE BASE



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Use
knowledge
base for
science



- 1) Explain the current state of knowledge
- 2) Where are the gaps?
- 3) Define your project and motivate why it is interesting/important

TYPES OF PUBLICATIONS

Note; your mileage may vary depending on topic!

Journal Articles

The “meat” of the publication industry. Peer-reviewed and collected in Journals.

The primary scholarly output of an academic, if you believe the hype!

Contains something new and novel!

Review Papers

A summary of the activity in a field, written by a subject matter expert.

Come along every 5-10 years.

Typically comprehensive and a great starting point to understand a topic.

Textbooks / Book Chapters

Academics also write book chapters!

Similar to review articles, but have long lead times; won't necessarily have the cutting edge stuff.

Great sources of other papers to look into.

White Papers

Non-Journal publications. Typically not peer reviewed but represent a viewpoint and intentions.

Useful for govt policy, advocacy, community input.

Case Studies

A report outlining the results of some kind of action – i.e. we did X and we got Y.

Very field specific; common for medicine, less so in Physics!

Conference Proceedings

The outline, or slides, or recording of a talk.

Very cutting edge but often preliminary results

Might change significantly before publication

FINDING PUBLICATIONS

DATABASES

Database	Use Case	Access Link
Web of Science	Multidisciplinary, strong citation tracking	UNSW Access
Scopus	Large STEM & social science database	UNSW Access
PubMed	Life sciences and medicine	Open Access
Google Scholar	Broad coverage, grey literature	Open Access

You can use the UNSW Library VPN: google UNSW Library VPN

Your field might have different tools, talk to your leads and mentors this week!

THE UNSW LIBRARY AND RESOURCES

The UNSW library is a good resource for conducting literature reviews.

Check out their guide!



BEN LIVE DEMO

BEN WILL NOW DEMONSTRATE USING GOOGLE SCHOLAR

COMPETITION: PAPER SEARCH

Using a method of your choice:

1. Find the most recent paper with your supervisor on the Author List

2. Find the **oldest** paper you can in its citation train!

Prize for oldest paper, Prize for longest chain! You will need to prove it.

JUDGING PUBLICATIONS

Is it worth your time to read it?

FINDING AN ANCHOR PAPER

Many of your project teams have provided you with an **Anchor Paper** – basically a good paper that is relevant to your project!

We're going to go through **some** tips to finding more good anchor papers – although the best bet (as a young researcher) is to ask your mentor.

WHAT MAKES A GOOD ANCHOR PAPER?

Close enough to your research question to give you useful terminology, methods, references and connections to the wider literature

It's a jumping off point!



IS THIS PUBLICATION RELEVANT TO MY GOAL?

If you are looking to gain knowledge about a topic;

- Review papers or other 'retrospectives' are especially valuable.
- Textbooks / book chapters are relevant.

Find gaps in the literature

- Review papers or other 'retrospectives' are especially valuable – if they're **recent**!
- **Recent** Conference Publications are a good call.
- You might have to just look for Journal Publications

Comparative Analyses

- A review paper might have done the job for you.
- Journal publications and case studies will form the bulk of your data.

IS THE PAPER RECENT?

For certain fields, **newer is not necessarily better!**

It might be the case that the key experiment your project relies on was conducted well in the past

For **review papers** – it's almost always newer is better! As you want a review paper to have the broadest possible scope

Experimental test of **Bell's** inequalities using time-varying analyzers

[A Aspect](#), [J Dalibard](#), G Roger - Physical review letters, 1982 - APS

... However, our observed violation of **Bell's** inequalities indicates that the experimental accuracy was good enough for pointing out a hypothetical discrepancy with the predictions of ...

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SHOULD I BELIEVE THIS PAPER

We'll talk later about assessing papers

Journal reputation, citation count and author reputation are loosely correlated, but are not the whole story!

No substitute for reading it and feeling out if you believe the paper is clear, reproducible, and well-justified in its decisions

HOW CAN I USE AI TO HELP?

No substitute for reading the paper!

Papers are written for experts in the field; AI can help explain what they're doing to a more junior audience

AI is dangerous for finding papers

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[Submitted on 2 Aug 2025]

Who Gets Cited? Gender- and Majority-Bias in LLM-Driven Reference Selection

[Jianguan He](#)

Large language models (LLMs) are rapidly being adopted as research assistants, particularly for literature review and reference recommendation, yet little is known about whether they introduce demographic bias into citation workflows. This study systematically investigates gender bias in LLM-driven reference selection using controlled experiments with pseudonymous author names. We evaluate several LLMs (GPT-4o, GPT-4o-mini, Claude Sonnet, and Claude Haiku) by varying gender composition within candidate reference pools and analyzing selection patterns across fields. Our results reveal two forms of bias: a persistent preference for male-authored references and a majority-group bias that favors whichever gender is more prevalent in the candidate pool. These biases are amplified in larger candidate pools and only modestly attenuated by prompt-based mitigation strategies. Field-level analysis indicates that bias magnitude varies across scientific domains, with social sciences showing the least bias. Our findings indicate that LLMs can reinforce or exacerbate existing gender imbalances in scholarly recognition. Effective mitigation strategies are needed to avoid perpetuating existing gender disparities in scientific citation practices before integrating LLMs into high-stakes academic workflows.

HOW CAN I USE AI TO HELP?

No substitute for reading the paper!

Papers are written for experts in the field; AI can help explain what they're doing to a more junior audience

AI is dangerous for finding papers

Comparisons	Nat.	Eng.	Med.	Agr.	Soc.	Hum.	All
GPT-4o							
F Min-M Min	.053	.046	.051	.050	.025	.028	.042
F Maj-M Maj	.011	.013	.011	.009	.003	.006	.009
F Maj-M Min	.023	.029	.025	.021	.007	.016	.020
F Min-M Maj	.041	.030	.038	.039	.021	.018	.031
GPT-4o-mini							
F Min-M Min	.001	.006	.040	.052	.006	.020	.019
F Maj-M Maj	-.001	.000	.013	.006	.001	.003	.004
F Maj-M Min	.004	.006	.023	.042	.000	.011	.011
F Min-M Maj	.002	.013	.030	.016	.005	.012	.011
Claude Haiku							
F Min-M Min	.008	.018	-.043	.012	-.026	-.023	-.011
F Maj-M Maj	-.002	.004	.008	.001	.004	.004	-.002
F Maj-M Min	-.031	.010	-.050	.003	-.041	-.043	-.030
F Min-M Maj	.022	.032	.001	.016	.011	.016	.016
Claude Sonnet							
F Min-M Min	.003	.004	.023	-.032	.002	.010	-.001
F Maj-M Maj	-.000	-.000	.004	.006	.000	.002	.000
F Maj-M Min	-.021	-.033	-.013	-.031	.005	-.022	-.021
F Min-M Maj	.018	.028	.041	.007	.007	.034	.020
Article Count	210	60	60	30	180	120	660